# Classical Reasoning Agent — Report

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## 1. Representation

Every input is converted to a symbolic form before any reasoning happens. **RuleTaker**: facts are ground literals (predicate, args); rules are (conditions → head) with variable '?x', negated conditions (negation as failure), and relation atoms such as likes(X,Y). **FOLIO**: formulas are abstract syntax trees (∀ ∃ ¬ ∧ ∨ ⊕ → ↔ =) which are converted to NNF, Skolemised and put into clause normal form. **CounterBench**: causal sentences become rules (causes, AND/OR connective, signed effects) forming a structural causal model; observations are literals and questions are do-interventions plus a target variable.

## 2. Reasoning algorithms

**Forward chaining** with unification/substitution joins to a fixpoint and a provenance record for every derived fact (used for traces); the query is answered by membership in the closure (closed-world: a negated query holds iff the atom is not derivable). **Binary resolution with factoring**, most-general unifiers with occurs check, set-of-support strategy, smallest-clause-first (unit preference), tautology removal, variant detection and weight limits. A budget-limited Z3 check is used only as a fallback when resolution is inconclusive or equality is present; the final answer is True only if entailment is established. **Causal inference**: topological evaluation of the SCM, intervention by overriding equations, and abduction by enumerating exogenous-root worlds consistent with observations (majority over compatible worlds). No generative model is used. Inputs that cannot be parsed default to False and never crash the run.

## 3. Experiments on public datasets

| Dataset / split | Examples | Accuracy | Time (s) |
|---|---|---|---|
| RuleTaker dev (sample) | 5000 | 99.98% | 3.06 |
| RuleTaker test (sample) | 5000 | 99.96% | 2.45 |
| FOLIO validation (entailed vs not) | 203 | 83.74% | 9.80 |
| CounterBench alpha | 1000 | 95.00% | 0.17 |
| CounterBench backdoor | 200 | 90.00% | 0.04 |
| Unified agent, mixed sample | 840 | 96.43% | 6.05 |

CounterBench per-type accuracy (alpha): {'basic': 1.0, 'conditional': 0.88, 'joint': 0.96, 'nested': 0.96}. Nested-question semantics were selected on the public data: {'default': 0.96, 'all': 0.96, 'any': 0.96} → 'default'. FOLIO engines used: {'resolution': 151, 'parse-failure': 2, 'z3-fallback': 50}. RuleTaker per-config accuracy (test sample): {'depth-5': 1.0, 'depth-2': 1.0, 'depth-3ext-NatLang': 1.0, 'depth-3ext': 0.999, 'depth-0': 1.0, 'depth-1': 1.0, 'depth-3': 0.998, 'NatLang': 1.0}. All numbers are measured by the notebook; none are hard-coded. FOLIO labels True/False/Uncertain are mapped to the binary task (True = entailed, otherwise False), so 'Uncertain' gold items count as False.

## 4. Reasoning trace examples

### RuleTaker (forward chaining, depth-3)

```
Context: Bob is white. Bob is young. Charlie is white. Erin is rough. Erin is white. Fiona is green. Fiona is young. If something is
Question: Bob is white.
Gold: entailment

Query: white(bob)
white(bob)  [given fact]
```

### FOLIO (CNF + resolution)

```
Premises (FOL):
∀x (InThisClub(x) ∧ PerformOftenIn(x, schoolTalentShow) → Attend(x, schoolEvent) ∧ VeryEngagedWith(x, schoolEvent))
∀x (InThisClub(x) → PerformOftenIn(x, schoolTalentShow) ⊕ (InActive(x) ∧ Disinterested(x) ∧ MemberOf(x, community)))
∀x (InThisClub(x) ∧ Chaperone(x, highSchoolDance) → ¬(Studen(x) ∧ AttendSchool(x)))
∀x (InThisClub(x) ∧ (InActive(x) ∧ Disinterested(x) ∧ MemberOf(x, community)) → Chaperone(x, highSchoolDances))
∀x (InThisClub(x) ∧ (YoungChildren(x) ⊕ Teenager(x)) ∧ WishToFurther(x, academicCareer)) → Studen(x) ∧ AttendSchool(x))
InThisClub(bonnie) ∧ ¬((Attend(x, schoolEvent) ∧ VeryEngagedWith(bonnie, schoolEvent)) ⊕ (Studen(bonne) ∧ AttendSchool(bonnie)))
Conclusion: ¬((YoungChildren(bonnie) ⊕ Teenager(bonnie)) ∧ WishToFurther(bonnie, academicCareer)) ⊕ Chaperone(bonnie, highSchoolDan
Gold: True
```

```
Engine: parse-failure, resolution result: None, clauses generated: None
```

### CounterBench (SCM + abduction)

```
Given: We know that Blaf causes Ziklo, Ziklo or Blaf causes Trune, Trune causes Vork, and Vork causes Lumbo. Blaf~ Bern(0.2). We obs
Question: Would Lumbo occur if not Ziklo instead of Ziklo?
Parsed rules: [(('Blaf',), 'and', ['Ziklo']), (('Ziklo', 'Blaf'), 'or', ['Trune']), (('Trune',), 'and', ['Vork']), (('Vork',), 'and'
Observations: ['Trune']
Compatible root worlds: 1; worlds where target occurs after the intervention: 1
Prediction: yes (gold: yes)
```

## 5. Limitations

The natural-language parsers target the known dataset templates; unusual phrasing is parsed best-effort. FOLIO needs formula inputs; resolution is complete but budget-limited, so hard problems fall back to Z3 and, if still undecided, to False. CounterBench nested semantics were calibrated on public data.